

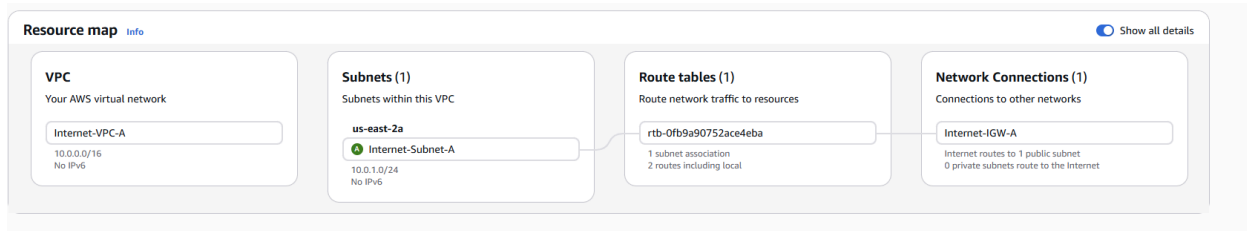
Task 1:

Region A Server Public IPv4: 3.145.108.228

- Us-east-2 region

Region B Client Public IPv4: 54.183.192.217

- Us-west-1 region

Step 1 Screenshot:**Step 2 Screenshot:**

Instance summary for i-066c3c34e99bc67ba (region-a-web-server) Info

Updated 15 minutes ago

Instance ID i-066c3c34e99bc67ba	Public IPv4 address 3.145.108.228 open address	Private IPv4 addresses 10.0.1.24
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-1-24.us-east-2.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-1-24.us-east-2.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 3.145.108.228 [Public IP]	VPC ID vpc-0f13335290b64ece5 (Internet-VPC-A)	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0fdc9d0b70adb533d (Internet-Subnet-A)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-2:865268032810:instance/i-066c3c34e99bc67ba	
Operator -		

```

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64
apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-core-2.4.66-1.amzn2023.0.1.x86_64
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_lua-2.4.66-1.amzn2023.0.1.x86_64
apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
httpd-2.4.66-1.amzn2023.0.1.x86_64
httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mod_http2-2.0.27-1.amzn2023.0.3.x86_64

Complete!
Hello from Region A server
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
Hello from Region A server
[ec2-user@ip-10-0-1-24 ~]$

```

Step 3 Screenshot:

Instance summary for i-0a7376fa27c583319 (region-b-client) info

Updated less than a minute ago

Instance ID

i-0a7376fa27c583319

IPv6 address

-

Hostname type

IP name: ip-10-1-1-241.us-west-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

54.183.192.217 [Public IP]

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

54.183.192.217 | open address

Instance state

Pending

Private IP DNS name (IPv4 only)

ip-10-1-1-241.us-west-1.compute.internal

Instance type

t2.micro

VPC ID

vpc-0772d6bb30a9b897b (Internet-VPC-B)

Subnet ID

subnet-0e34eef37e62ca77e (Internet-vpc-b-subnet)

Instance ARN

arn:aws:ec2:us-west-1:865268032810:instance/i-0a7376fa27c583319

Private IPv4 addresses

10.1.1.241

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

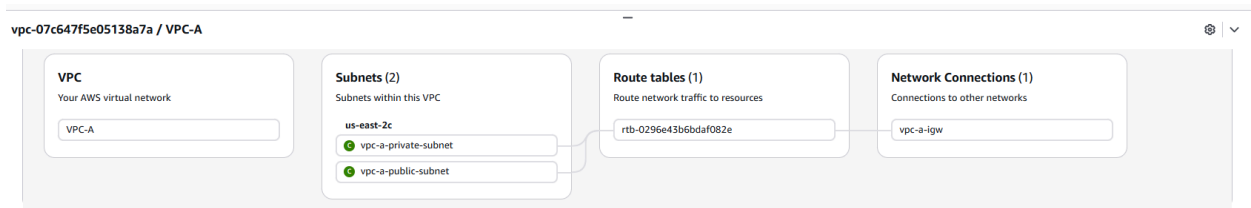
false

Step 4 Screenshot:

```
[ec2-user@ip-10-1-1-241 ~]$ curl http://3.145.108.228
Hello from Region A server
[ec2-user@ip-10-1-1-241 ~]$
```

Task 2:

Step 1 Screenshots:



Step 2 Screenshots:

Instance summary for i-0e1903e849d811be1 (task2-private-server) info

Updated less than a minute ago

Instance ID

i-0e1903e849d811be1

IPv6 address

-

Hostname type

IP name: ip-10-10-2-137.us-east-2.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

-

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-10-2-137.us-east-2.compute.internal

Instance type

t2.micro

VPC ID

vpc-07c647f5e05138a7a (VPC-A)

Subnet ID

subnet-0e622f66427d97435 (vpc-a-private-subnet)

Instance ARN

arn:aws:ec2:us-east-2:865268032810:instance/i-0e1903e849d811be1

Private IPv4 addresses

10.10.2.137

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance details

```
Complete!
Hello from PrivateLink server
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
Hello from PrivateLink server
[ec2-user@ip-10-10-2-137 ~]$
```

Step 3 Screenshots:

EC2 > Load balancers > task2-nlb

Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager [New](#)

▼ **Images**
AMIs
AMI Catalog

▼ **Elastic Block Store**
Volumes
Snapshots
Lifecycle Manager

▼ **Network & Security**
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ **Load Balancing**
Load Balancers
Target Groups
Trust Stores

▼ **Auto Scaling**
Auto Scaling Groups

Settings

Introducing configurable log deliveries for NLB
Send Network Load Balancer logs to multiple destinations — CloudWatch, S3, or Kinesis Firehose — giving you greater flexibility in how you store, analyze, and manage your load balancer logs. [Learn more](#)

task2-nlb

Actions

▼ **Details**

Load balancer type Network	Status Active	VPC vpc-07c647f5e05138a7a	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZLMOA37VPKANP	Availability Zones subnet-0c993e6b9474f7807e us-east-2c (use2-az3)	Date created March 24, 2026, 20:38 (UTC-05:00)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-2:865268032810:loadbalancer/net/task2-nlb/21c75c899476d7d8		DNS name info task2-nlb-21c75c899476d7d8.elb.us-east-2.amazonaws.com (A Record)	

Listeners | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

Listeners (1)
A listener checks for connection requests using the protocol and port that you configure. Traffic received by a Network Load Balancer listener is forwarded to the selected target group.

Filter listeners

☐	Protocol/Port	Default action	ARN	Security policy	Default SSL/TLS certificate	ALPN policy	Tags
☐	TCP:80	Forward to target group task2-private-tg 1 (100%) Target group stickiness: Off	ARN	Not applicable	Not applicable	None	0 tags

EC2 > Target groups

Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager [New](#)

▼ **Images**
AMIs
AMI Catalog

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Volumes
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Lifecycle Manager

▼ **Network & Security**
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Placement Groups
Key Pairs
Network Interfaces

▼ **Load Balancing**
Load Balancers
Target Groups
Trust Stores

▼ **Auto Scaling**
Auto Scaling Groups

Target groups (1/2) [Info](#) | [What's new?](#)

Filter target groups

☐	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
☐	A2-L1	arn:aws:elasticloadbalancing:us-east-2:865268032810:targetgroup/A2-L1/21c75c899476d7d8	8443	TCP	Instance	None associated	vpc-0a7bf477be5a3a626
☒	task2-private-tg	arn:aws:elasticloadbalancing:us-east-2:865268032810:targetgroup/task2-private-tg/21c75c899476d7d8	80	TCP	Instance	task2-nlb	vpc-07c647f5e05138a7a

Target group: task2-private-tg

Details | **Targets** | Monitoring | Health checks | Attributes | Tags

Registered targets (1)
Filter targets

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Administrative o...	Override details	Launch time
☐	i-0e1903e849d811...	task2-private-server	80	us-east-2c (use2-az3)	Healthy		No override	No override is curre...	March 24, 2026, 18...

Step 4 Screenshot:

aws

EC2 > Endpoint services > vpc-svc-077ab35430a5477a3

DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

▼ **Security**
Network ACLs
Security groups

▼ **PrivateLink and Lattice**
Getting started
Endpoints
Endpoint services
Service networks
Lattice services
Resource configurations
Resource gateways
Target groups
Domain verifications [New](#)

▼ **DNS firewall**
Rule groups
Domain lists

You have successfully created endpoint service configuration for vpc-svc-077ab35430a5477a3 / VPC-B.

vpc-svc-077ab35430a5477a3 / VPC-B

Actions

Details

Service ID vpc-svc-077ab35430a5477a3	Types Interface	State Available
Network Load Balancers ARNs arn:aws:elasticloadbalancing:us-east-2:865268032810:loadbalancer/net/task2-nlb/21c75c899476d7d8	Gateway Load Balancers ARNs -	Acceptance required Yes
DNS names vpc-svc-077ab35430a5477a3.us-east-2.vpc.amazonaws.com	Remote Access Enabled false	Domain verification status info
Domain verification name -	Domain verification type info	Supported IP address type IPv4

Load balancers | Allow principals | Endpoint connections | Notifications | Supported Regions | Monitoring | Contributor Insights | Tags

Load balancers (1) [Info](#)

Search

Availability Zone	Load balancer names	State	ARNs
use2-az3 (us-east-2c)	task2-nlb	Active	1 ARN

Step 5 Screenshot:

vpce-0a0c6897340e4caf2 / vpc-b-endpoint Actions ▾

Details

Endpoint ID vpce-0a0c6897340e4caf2	Status Available
VPC ID vpc-0d5c0fee4b1e0dcb0 (VPC-B)	Status message -
DNS record IP type ipv4	IP address type ipv4
Private DNS preference -	Private DNS specified domains -
Creation time Wednesday, March 25, 2026 at 13:10:33 CDT	Endpoint type Interface
Service name com.amazonaws.vpce.us-east-2.vpce-svc-077ab35430a5477a3	Private DNS names enabled No
Service region us-east-2	DNS names vpce-0a0c6897340e4caf2-3tfphatr.vpce-svc-077ab35430a5477a3.us-east-2.vpce.amazonaws.com- (ZC8PG0KIFKBRI) vpce-0a0c6897340e4caf2-3tfphatr-us-east-2.vpce-svc-077ab35430a5477a3.us-east-2.vpce.amazonaws.com- (ZC8PG0KIFKBRI)

Step 6 Screenshot:

```
[ec2-user@ip-10-20-1-166 ~]$ curl http://vpce-0a0c6897340e4caf2-3tfphatr.vpce-svc-077ab35430a5477a3.us-east-2.vpce.amazonaws.com
Hello from PrivateLink server
[ec2-user@ip-10-20-1-166 ~]$
```

bastion public IP: 18.222.153.65

private server private IP: 10.10.2.137

client public IP: 3.15.220.250

client private IP: 10.20.1.166

endpoint DNS:

vpce-0a0c6897340e4caf2-3tfphatr.vpce-svc-077ab35430a5477a3.us-east-2.vpce.amazonaws.com

Task 3:

Step 1 Screenshots:

vpc-021e75a7acaa6185a / VPC-1 Actions ▾

Details Info

VPC ID vpc-021e75a7acaa6185a	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-060d536497f526bf0	Main route table rtb-0b7fe69984e48d0c9
Main network ACL acl-015a8208a4d8b230b	Default VPC No	IPv4 CIDR 10.30.0.0/16	IPv6 pool -
IPv6 CIDR -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 865268032810
Encryption control ID -	Encryption control mode -		

Resource map Info

VPC
Your AWS virtual network
VPC-1

Subnets (2)
Subnets within this VPC
us-east-2c
vpc-1-public-subnet
vpc-1-private-subnet

Route tables (3)
Route network traffic to resources
vpc-1-rt-private
vpc-1-rt-public
rtb-0b7fe69984e48d0c9

Network Connections (1)
Connections to other networks
vpc-1-igw

vpc-0375cb7d7e9bd2622 / VPC-2

Actions

Details

Info

VPC ID

vpc-0375cb7d7e9bd2622

DNS resolution

Enabled

Main network ACL

acl-0d5f803b6fc0a8b24

IPv6 CIDR

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-060d536497f526bf0

IPv4 CIDR

10.40.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-0bb40981cf8711dff

IPv6 pool

-

Owner ID

865268032810

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

Info

Show all details

VPC

Your AWS virtual network

VPC-2

Subnets (2)

Subnets within this VPC

us-east-2c

vpc-2-public-subnet

vpc-2-private-subnet

Route tables (3)

Route network traffic to resources

vpc-2-rt-public

vpc-2-rt-private

rtb-0bb40981cf8711dff

Network Connections (1)

Connections to other networks

vpc-2-igw

Step 2 Screenshots:

Instance summary for i-0455752bd88f6d840 (task3-server)

Info

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID

i-0455752bd88f6d840

IPv6 address

-

Hostname type

IP name: ip-10-30-2-7.us-east-2.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

-

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-30-2-7.us-east-2.compute.internal

Instance type

t2.micro

VPC ID

vpc-021e75a7acaa6185a (VPC-1)

Subnet ID

subnet-0978f99e2e3554b57 (vpc-1-private-subnet)

Instance ARN

arn:aws:ec2:us-east-2:865268032810:instance/i-0455752bd88f6d840

Private IPv4 addresses

10.30.2.7

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance details

Info

AMI ID

ami-0b0b78dcacbab728f

AMI name

al2023-ami-2023.10.20260302.1-kernel-6.1-x86_64

Stop protection

Disabled

Monitoring

disabled

Allowed image

-

Launch time

Wed Mar 25 2026 14:44:39 GMT-0500 (Central Daylight Time) (29 minutes)

Platform details

Linux/UNIX

Termination protection

Disabled

AMI location

amazon/al2023-ami-2023.10.20260302.1-kernel-6.1-x86_64

Instance summary for i-065412d36133715a4 (task3-client)

Info

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID

i-065412d36133715a4

IPv6 address

-

Hostname type

IP name: ip-10-40-1-42.us-east-2.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

18.117.112.228 [Public IP]

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

18.117.112.228 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-40-1-42.us-east-2.compute.internal

Instance type

t2.micro

VPC ID

vpc-0375cb7d7e9bd2622 (VPC-2)

Subnet ID

subnet-03b9f11a22a60b880 (vpc-2-public-subnet)

Instance ARN

arn:aws:ec2:us-east-2:865268032810:instance/i-065412d36133715a4

Private IPv4 addresses

10.40.1.42

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance details

Info

AMI ID

ami-0b0b78dcacbab728f

AMI name

al2023-ami-2023.10.20260302.1-kernel-6.1-x86_64

Stop protection

Disabled

Monitoring

disabled

Allowed image

-

Launch time

Wed Mar 25 2026 14:47:30 GMT-0500 (Central Daylight Time) (about 1 hour)

Platform details

Linux/UNIX

Termination protection

Disabled

AMI location

amazon/al2023-ami-2023.10.20260302.1-kernel-6.1-x86_64

Instance summary for i-018f8cbb4697bc8d (task3-bastion) Info

Updated less than a minute ago

Instance ID

i-018f8cbb4697bc8d

IPv6 address

–

Hostname type

IP name: ip-10-30-1-4.us-east-2.compute.internal

Answer private resource DNS name

–

Auto-assigned IP address

13.59.230.142 [Public IP]

IAM role

–

IMDSv2

Required

Operator

–

Public IPv4 address

13.59.230.142 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-30-1-4.us-east-2.compute.internal

Instance type

t2.micro

VPC ID

vpc-021e75a7acaa6185a (VPC-1)

Subnet ID

subnet-0386ee96ede39400f (vpc-1-public-subnet)

Instance ARN

arn:aws:ec2:us-east-2:865268032810:instance/i-018f8cbb4697bc8d

Private IPv4 addresses

10.30.1.4

Public DNS

–

Elastic IP addresses

–

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

–

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance details Info

AMI ID

ami-0b0b78dcacbab728f

AMI name

al2023-ami-2023.10.20260302.1-kernel-6.1-x86_64

Stop protection

Disabled

Monitoring

disabled

Allowed image

–

Launch time

Wed Mar 25 2026 14:40:37 GMT-0500 (Central Daylight Time) (about 1 hour)

Platform details

Linux/UNIX

Termination protection

Disabled

AMI location

amazon/al2023-ami-2023.10.20260302.1-kernel-6.1-x86_64

```
Complete!
Hello from Peering server
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
Hello from Peering server
[ec2-user@ip-10-30-2-7 ~]$ |
```

Step 3 Screenshot:

pcx-09491cbe21fad0262 / vpc-peering-connection-1 Actions

Details Info

Requester owner ID

865268032810

Peering connection ID

pcx-09491cbe21fad0262

Status

Active

Expiration time

–

Accepter owner ID

865268032810

Requester VPC

vpc-021e75a7acaa6185a / VPC-1

Requester CIDRs

10.30.0.0/16

Requester Region

Ohio (us-east-2)

VPC Peering connection ARN

arn:aws:ec2:us-east-2:865268032810:vpc-peering-connection/pcx-09491cbe21fad0262

Accepter VPC

vpc-0375cb7d7e9bd2622 / VPC-2

Accepter CIDRs

10.40.0.0/16

Accepter Region

Ohio (us-east-2)

DNS

Route tables

Tags

DNS settings Edit DNS settings

Requester VPC (vpc-021e75a7acaa6185a / VPC-1) Info

Allow accepter VPC to resolve DNS of hosts in requester VPC to private IP addresses

Disabled

Accepter VPC (vpc-0375cb7d7e9bd2622 / VPC-2) Info

Allow requester VPC to resolve DNS of hosts in accepter VPC to private IP addresses

Disabled

Step 4 Screenshots:

rtb-0e34fa91bca5e745c / vpc-2-rt-public

DetailsInfo

Route table ID

rtb-0e34fa91bca5e745c

VPC

vpc-0375cb7d7e9bd2622 | VPC-2

Main

No

Owner ID

865268032810

Explicit subnet associations

subnet-03b9f11a22a60b880 / vpc-2-public-subnet

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (3)

BothEdit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0229b2...	Active	No	Create Route
10.30.0.0/16	pcx-09491cb...	Active	No	Create Route
10.40.0.0/16	local	Active	No	Create Route Table

rtb-07e3ccd37ada8aab5 / vpc-1-private-rt

DetailsInfo

Route table ID

rtb-07e3ccd37ada8aab5

VPC

vpc-021e757acaa6185a | VPC-1

Main

No

Owner ID

865268032810

Explicit subnet associations

subnet-0978f99e2e3554b57 / vpc-1-private-subnet

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (3)

BothEdit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-006214b5c556a5641	Active	No	Create Route
10.30.0.0/16	local	Active	No	Create Route Table
10.40.0.0/16	pcx-09491cbe21fad0262	Active	No	Create Route